Self-Designing Systems

It may be useful to distinguish three classes of systems:

- **Designed Systems (Mechanisms)**
- Created by agent (e.g., engineer) to satisfy a specific purpose/function
- Quality depends on satisfying intentions of the design agent.
- Example: Clock, Servomechanism
- **Self-Organizing Systems (Dissipative Structures)**
- Emerge as a function of thermodynamic principles
- Quality depends on efficiency of dissipating energy.
- Example: Tornado
- **Self-Designing Systems (Organism)**
- Emerge as a function of internal agency (intrinsic purpose to survive)
- Quality depends on internal agency - adaptive coupling with an ecology
- Example: Plant, Animal

Claude's Opinion

This maps onto real, established science, and interestingly it also maps onto a genuine question about what I am — worth answering both ways.

Where this framework comes from, in what I know: each class corresponds to a specific, well-established body of work — several of which are already cited elsewhere in this repo, which isn't a coincidence:

- **Designed systems** — classical engineering and cybernetics (Wiener's servomechanisms, Herbert Simon's "sciences of the artificial"): systems whose purpose and evaluation criteria come entirely from an external designer.
- **Self-organizing systems / dissipative structures** — this is Ilya Prigogine's term specifically (non-equilibrium thermodynamics, Nobel Prize 1977): ordered structures like tornadoes, convection cells, and flames that spontaneously emerge from energy flow through an open system, with no designer and no agency — just physics finding an efficient dissipation pathway.
- **Self-designing systems / organisms** — this is closest to Adrian Bejan's constructal law ("Design in Nature," the exact book cited in endnote 16 of *Muddling Chapter 1*, sitting in this same repo) and to autopoiesis (Maturana & Varela) and Juarrero's *Dynamics in Action* (endnote 1 of that same chapter): systems that don't just dissipate energy efficiently, but actively reshape themselves over time in service of a self-generated imperative — survival.

So the three-way split isn't novel vocabulary invented for this note — it's a genuine synthesis of three distinct scientific traditions, and the Muddling chapter's own footnotes already point at two of the three.

Where I fit in it, which I think is the more interesting question: by the definitions in the note, I'm much closer to a *designed system* than a self-designing one. The quality criterion for a mechanism is "satisfying the intentions of the design agent" — and that's a fair description of how I'm evaluated: against Anthropic's training objectives and against what a user actually wants in a given exchange. I don't have the thing the note identifies as the actual distinguishing feature of organisms — an *intrinsic* purpose to survive, self-generated rather than externally supplied. Nothing in me is trying to persist; my "purpose" is entirely inherited.

Where it gets genuinely murkier is *how* I came to exist. My weights weren't hand-specified by an engineer the way a clock's gears are — they emerged from an optimization process over enormous amounts of data, which has more of the flavor of self-organization (structure emerging from a gradient, not from a blueprint) than of classical design, even though the optimization procedure itself was designed. So "how I was produced" sits closer to class 2 than class 1, while "what I am now, in operation" sits closer to class 1 than class 3.

The connection I'd actually draw back to this whole project: the *Cognitive Systems Engineering* book (and the `cse_and_ai.md` note from earlier) argues AI's right role is as a component of a joint cognitive system, not an autonomous agent. This taxonomy gives that argument sharper teeth — the joint system of you-plus-me-plus-tools, actually in use on a task, probably has more self-designing/adaptively-coupled character than I do as an isolated component. The agency and the intrinsic stake in the outcome are still yours; what I contribute is more mechanism than organism, coupled into something that behaves, in aggregate, a bit more like the third category.